

Usage of gender-neutral pronouns in English literature

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Abstract

Gender-neutral pronouns can be used to represent non-binary people. Pronouns in English literature have mostly followed binary gender conventions. The usage of pronouns in literature can be influenced by the view our society has on genders. Books from the Project Gutenberg corpus are used to search for gendered and gender-neutral pronouns throughout history. An increase in gender-neutral pronouns in the more recent book suggests that the usage of gendered and neutral pronouns has increased over time in English literature.

1 Introduction

The use of gender-neutral pronouns has become an important topic of discussion. During the last century, society has begun to recognize other gender identities than the traditional binary male and female. Pronouns in literature have mostly followed binary gender conventions, using "he" and "she" to represent male and female characters. However, as awareness of non-binary, genderqueer, and transgender individuals has grown in recent years, so too has the need for language that includes more people and their gender identities. Many people do not feel like they fit into the male/female categories and wish to be addressed using gender-neutral language.

The usage of gendered pronouns can be influenced by the view our society has on genders. The way authors depict these genders in various literature can be representative of the general public's view of genders. English books often only used male or female pronouns in the past. However, in recent years there has been a focus on gender-neutral pronouns. These changes will be studied

through the Project Gutenberg corpus. If there has been a significant shift in the usage of these pronouns, it may provide insight into society's attitudes towards genders and how they change over time.

This paper studies the usage of gender-neutral pronouns in literature throughout the years, namely the relative amount of gender-neutral pronouns used compared to gendered pronouns. Literature is always changing throughout the years, and English literature is no different. So has the usage of gendered and neutral pronouns changed over time in English literature?

2 Related Work

Sendén M. G. (2021) studies the changes in attitudes toward the Swedish gender-inclusive pronoun "hen" in 2015 and 2018. They have conducted an experiment where they collect data from the Swedish population. According to them,

"Both attitudes and use changed significantly between these years, such that attitudes became more positive and use of hen increased."

This could indicate that the public attitude towards gender-neutral pronouns may be improving in recent years, at least in the Swedish population. The study, however, only gathers data from surveys, not from any English literature. It is also mentioned that there are many other factors that could have influenced the surveys, such as the Swedish "hen" being both a generic pronoun and a non-binary pronoun, or "hen" being officially recognized as a word.

Pronoun usage in books seems to align with the status for a demographic (Twenge J.M., 2012). Gendered pronoun usage in U.S. books is studied alongside women's status over the last hundred years. Women's status is quantified using

the number of degrees awarded to women. They found that there is a link between pronoun usage in books and women's status. The data for pronoun use is gathered from Google Books, and only U.S. books were used. This has limited their results on language use and women's status to the U.S. only. Also, there was no focus on gender-neutral pronouns, the aim of the study was male and female pronouns.

Noll J. (2018) tries to determine whether "they" or "he" is more effective as epicene pronouns (pronouns used for unknown or unspecified genders). Participants were asked if they agreed or disagreed with certain phrases that were presented to them. The phrases contain either gendered or gender-neutral probe words. The response time with which the participants agree or disagree is noted. Longer response times mean that the participants are less inclusive of the category of the probe words. Noll observes that "they" is more inclusive than "he". "he" is seen as a more masculine than feminine epicene pronoun compared to "they". It is acknowledged that assertions about differences in responses over time were not directly assessed.¹

Using these studies in mind, a hypothesis may be constructed. Gender-neutral pronouns are becoming more favorable in the public's eyes, pronoun usage in books seems to reflect status. The usage of gender-neutral pronouns is increasing over time compared to gendered pronouns in English literature.

3 Data

Books from the Project Gutenberg corpus will be used as data for this study. The corpus contains mostly English literature from Western culture in the form of eBooks. Project Gutenberg only accepts books in the public domain, and no permission is needed for non-commercial use. The corpus includes many books from the 19th and 20th centuries, and almost no books from the 21st century. The goal of this paper is to study the difference in pronouns between time periods, so books from the 19th century will be compared to books from the 20th century. The Project Gutenberg corpus is a great fit for this, as it has the most popular books for the time periods we are comparing. This should be relatively representative of the general language used in that time period. For simplicity's

sake, only one book will be chosen per time period.

The books are available to download in many file formats, including EPUB and plain text files. We will be using the plain text files with one small modification. Any text added by Project Gutenberg will be removed. The rest of the text will be kept as is. This step could be automated, as the lines indicating the added sections are the only lines containing three consecutive asterisks ("***").

The file will be searched for occurrences of pronouns using a grep command in Linux. The file will be tokenized per word, then sorted, counted, and filtered on the pronouns we need. First only the gender-neutral pronouns are counted, then the total number of pronouns. This way we can calculate the percentage of pronouns that are gender-neutral, alongside the percentage of gendered pronouns. We then construct a table with these percentages.

	Book 1	Book 2
Gender-neutral	x%	y%
Gendered	(100 - x)%	(100 - y)%

Table 1: Example of a table of percentages of gender-neutral and gendered pronouns in two books

The independent variable here is the year it was published in, and the dependent variable is the percentage of gender-neutral pronouns. Book 1 was published in the 19th century, and book 2 was published in the 20th century. The following is an overview of the pronouns that were searched for:

- Gendered: he, she, him, her, his, her, hers, himself, herself.
- Gender-neutral: they, them, their, theirs, themselves.

Other pronouns are not included as they are not used in the same contexts as the ones above.

4 Predicted Results

These are rough estimates of the percentages of gender-neutral and gendered pronouns in the two books. The numbers themselves are not the focus. The important part here is the increase in percentages from the 19th century to the 20th century. We predict that gender-neutral pronoun usage will increase with time, as previous literature

¹[Two experiments were done, 15 years apart.]

	19th century book	20th century book
Gender-neutral	30%	40%
Gendered	70%	60%

Table 2: Estimated percentages of gender-neutral and gendered pronouns in two books

seems to suggest so. The exact percentage from actual books will likely differ from the values in the table, but we predict that they will be increasing with time.

5 Conclusion

An increase in gender-neutral pronouns may indicate that many authors try to use more gender-neutral terms in their books. Literature changes with the culture of the authors, and are representative of the public's attitude towards gender-neutral pronouns. The usage of gendered and neutral pronouns has increased over time in English literature. The general public's attitude towards gender-neutral pronouns have thus improved as well.

Ideally we would include many more books from the Project Gutenberg corpus and plot their percentages against their year of publication. A trend may be more visible in a graph. Further research may include using another corpus, such as Google Books, to search through a bigger time period. The ratio between male, female and gender-neutral pronouns may also be recorded.

[Link to Github repository](#)

References

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